

General

Detalles

Emitido para

Nombre común (CN)	www.una.ac.cr
Organización (O)	<Not Part Of Certificate>
Unidad organizativa (OU)	<Not Part Of Certificate>

Emitido por

Nombre común (CN)	E7
Organización (O)	Let's Encrypt
Unidad organizativa (OU)	<Not Part Of Certificate>

Período de validez

Emitido el	lunes, 13 de abril de 2026, 2:03:50 p.m.
Expira el	domingo, 12 de julio de 2026, 2:03:49 p.m.

Huellas digitales de SHA-256

Certificado	aeb7fd6391431229189c13a65d8fe0810c2ddfb0464338495b989638dadbf99
Clave pública	94289eddcf1b41311c4e88ed2ec60cecb8efaafbca8afc41060559471d86ec66

```
jarod@Jarod:/mnt/c/Users/jarod/Documents/Tercer año/l semestre/S.O/linux$ echo | openssl s_client -connect www.una.ac.cr:443 -servername www.una.ac.cr 2>/dev/null | openssl x509 -outform PEM > una.crt

openssl x509 -in una.crt -noout -subject
openssl x509 -in una.crt -noout -issuer
openssl x509 -in una.crt -noout -dates
openssl x509 -in una.crt -noout -ext subjectAltName
openssl x509 -in una.crt -noout -fingerprint -sha256
subject=CN = www.una.ac.cr
issuer=C = US, O = Let's Encrypt, CN = E7
notBefore=Apr 13 20:03:50 2026 GMT
notAfter=Jul 12 20:03:49 2026 GMT
X509v3 Subject Alternative Name:
    DNS:www.una.ac.cr
sha256 Fingerprint=AE:B7:FD:63:91:43:12:29:18:9C:13:A6:5D:8F:E0:81:0C:2D:DF:B0:46:43:38:49:5B:98:96:38:DA:DB:FF:99
```

```
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: ECDSA
Server Temp Key: ECDH, prime256v1, 256 bits
---
SSL handshake has read 2344 bytes and written 441 bytes
Verification: OK
---
New, TLSv1.2, Cipher is ECDHE-ECDSA-AES128-GCM-SHA256
Server public key is 256 bit
Secure Renegotiation IS supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
SSL-Session:
    Protocol : TLSv1.2
    Cipher   : ECDHE-ECDSA-AES128-GCM-SHA256
    Session-ID: E2D86472371DD856CBA109739C6580D93A2F76C0DD2D9424ED9DFFE16EF5D0EF
    Session-ID-ctx:
    Master-Key: E34D58E848729AF24884A658B37A0C0C2D0076F55256DD29C802ECCC658473D90D22A4A71AAEBE42C54275041B9016E4
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    Start Time: 1780719207
    Timeout    : 7200 (sec)
    Verify return code: 0 (ok)
    Extended master secret: yes
---
jarod@Jarod:/mnt/c/Users/jarod/Documents/Tercer año/l semestre/S.O/linux$
```

```
jarod@Jarod:/mnt/c/Users/jarod/Documents/Tercer año/l semestre/S.O/linux$ openssl genrsa -out miCA.key 4096
openssl req -x509 -new -nodes -key miCA.key -sha256 -days 365 -out miCA.crt -subj "/C=CR/O=Laboratorio Redes/OU=Segurida
d/CN=Mi CA Local"
openssl genrsa -out servidor.key 2048
# (crear servidor.cnf como indica la sección 12.2)
openssl req -new -key servidor.key -out servidor.csr -config servidor.cnf
openssl x509 -req -in servidor.csr -CA miCA.crt -CAkey miCA.key -CAcreateserial -out servidor.crt -days 90 -sha256 -extf
ile servidor.cnf -extensions v3_req
Can't open "servidor.cnf" for reading, No such file or directory
40076852027D0000:error:80000002:system library:BIO_new_file:No such file or directory:../crypto/bio/bss_file.c:67:callin
g fopen(servidor.cnf, r)
40076852027D0000:error:10000080:BIO routines:BIO_new_file:no such file:../crypto/bio/bss_file.c:75:
Can't open "servidor.cnf" for reading, No such file or directory
403738F3E4790000:error:80000002:system library:BIO_new_file:No such file or directory:../crypto/bio/bss_file.c:67:callin
g fopen(servidor.cnf, r)
403738F3E4790000:error:10000080:BIO routines:BIO_new_file:no such file:../crypto/bio/bss_file.c:75:
jarod@Jarod:/mnt/c/Users/jarod/Documents/Tercer año/l semestre/S.O/linux$
```

```
jarod@Jarod:/mnt/c/Users/jarod/Documents/Tercer año/l semestre/S.O/linux$ openssl s_server -cert servidor.crt -key servi
dor.key -accept 8443 -www
Using default temp DH parameters
ACCEPT
```

```
s_server -cert servidor.crt -key servidor.key -accept 8443 -www
```

```
Secure Renegotiation IS NOT supported
```

```
Ciphers supported in s_server binary
```

TLSv1.3	:TLS_AES_256_GCM_SHA384	TLSv1.3	:TLS_CHACHA20_POLY1305_SHA256
TLSv1.3	:TLS_AES_128_GCM_SHA256	TLSv1.2	:ECDHE-ECDSA-AES256-GCM-SHA384
TLSv1.2	:ECDHE-RSA-AES256-GCM-SHA384	TLSv1.2	:DHE-RSA-AES256-GCM-SHA384
TLSv1.2	:ECDHE-ECDSA-CHACHA20-POLY1305	TLSv1.2	:ECDHE-RSA-CHACHA20-POLY1305
TLSv1.2	:DHE-RSA-CHACHA20-POLY1305	TLSv1.2	:ECDHE-ECDSA-AES128-GCM-SHA256
TLSv1.2	:ECDHE-RSA-AES128-GCM-SHA256	TLSv1.2	:DHE-RSA-AES128-GCM-SHA256
TLSv1.2	:ECDHE-ECDSA-AES256-SHA384	TLSv1.2	:ECDHE-RSA-AES256-SHA384
TLSv1.2	:DHE-RSA-AES256-SHA256	TLSv1.2	:ECDHE-ECDSA-AES128-SHA256
TLSv1.2	:ECDHE-RSA-AES128-SHA256	TLSv1.2	:DHE-RSA-AES128-SHA256
TLSv1.0	:ECDHE-ECDSA-AES256-SHA	TLSv1.0	:ECDHE-RSA-AES256-SHA
SSLv3	:DHE-RSA-AES256-SHA	TLSv1.0	:ECDHE-ECDSA-AES128-SHA
TLSv1.0	:ECDHE-RSA-AES128-SHA	SSLv3	:DHE-RSA-AES128-SHA
TLSv1.2	:RSA-PSK-AES256-GCM-SHA384	TLSv1.2	:DHE-PSK-AES256-GCM-SHA384
TLSv1.2	:RSA-PSK-CHACHA20-POLY1305	TLSv1.2	:DHE-PSK-CHACHA20-POLY1305
TLSv1.2	:ECDHE-PSK-CHACHA20-POLY1305	TLSv1.2	:AES256-GCM-SHA384
TLSv1.2	:PSK-AES256-GCM-SHA384	TLSv1.2	:PSK-CHACHA20-POLY1305
TLSv1.2	:RSA-PSK-AES128-GCM-SHA256	TLSv1.2	:DHE-PSK-AES128-GCM-SHA256
TLSv1.2	:AES128-GCM-SHA256	TLSv1.2	:PSK-AES128-GCM-SHA256
TLSv1.2	:AES256-SHA256	TLSv1.2	:AES128-SHA256
TLSv1.0	:ECDHE-PSK-AES256-CBC-SHA384	TLSv1.0	:ECDHE-PSK-AES256-CBC-SHA
SSLv3	:SRP-RSA-AES-256-CBC-SHA	SSLv3	:SRP-AES-256-CBC-SHA
TLSv1.0	:RSA-PSK-AES256-CBC-SHA384	TLSv1.0	:DHE-PSK-AES256-CBC-SHA384
SSLv3	:RSA-PSK-AES256-CBC-SHA	SSLv3	:DHE-PSK-AES256-CBC-SHA
SSLv3	:AES256-SHA	TLSv1.0	:PSK-AES256-CBC-SHA384
SSLv3	:PSK-AES256-CBC-SHA	TLSv1.0	:ECDHE-PSK-AES128-CBC-SHA256
TLSv1.0	:ECDHE-PSK-AES128-CBC-SHA	SSLv3	:SRP-RSA-AES-128-CBC-SHA
SSLv3	:SRP-AES-128-CBC-SHA	TLSv1.0	:RSA-PSK-AES128-CBC-SHA256
TLSv1.0	:DHE-PSK-AES128-CBC-SHA256	SSLv3	:RSA-PSK-AES128-CBC-SHA
SSLv3	:DHE-PSK-AES128-CBC-SHA	SSLv3	:AES128-SHA
TLSv1.0	:PSK-AES128-CBC-SHA256	SSLv3	:PSK-AES128-CBC-SHA

```
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```

```
Ciphers common between both SSL end points:
```

TLS_AES_128_GCM_SHA256	TLS_AES_256_GCM_SHA384	TLS_CHACHA20_POLY1305_SHA256
ECDHE-ECDSA-AES128-GCM-SHA256	ECDHE-RSA-AES128-GCM-SHA256	ECDHE-ECDSA-AES256-GCM-SHA384
ECDHE-RSA-AES256-GCM-SHA384	ECDHE-ECDSA-CHACHA20-POLY1305	ECDHE-RSA-CHACHA20-POLY1305
ECDHE-RSA-AES128-SHA	ECDHE-RSA-AES256-SHA	AES128-GCM-SHA256
AES256-GCM-SHA384	AES128-SHA	AES256-SHA

```
Signature Algorithms: ECDSA+SHA256:RSA-PSS+SHA256:RSA+SHA256:ECDSA+SHA384:RSA-PSS+SHA384:RSA+SHA384:RSA-PSS+SHA512:RSA+SHA512
```

```
Shared Signature Algorithms: ECDSA+SHA256:RSA-PSS+SHA256:RSA+SHA256:ECDSA+SHA384:RSA-PSS+SHA384:RSA+SHA384:RSA-PSS+SHA512:RSA+SHA512
```

```
Supported groups: ::x25519:secp256r1:secp384r1
```

```
Shared groups: x25519:secp256r1:secp384r1
```

```
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```

```
New, TLSv1.3, Cipher is TLS_AES_128_GCM_SHA256
```

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inglés?

Traducir a

español